UKRI Living Benchmarks NEMO

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1 Introduction

This report describes how to build and run the NEMO ocean model using both the Spack package manager and provided scripts for manual compilation. It covers the required dependencies, compiler and the general setup needed on HPC systems. The report also includes details on building and running NEMO on GPUs through PSyclone [2], along with any limitations and practical considerations.

Performance results are presented for the ORCA1, ORCA025, ORCA12, and ORCA36 resolution models on Archer 2, Isambard 3, and Isambard-AI supercomputers. The GPU results show that NEMO is now capable of running on Nvidia GPUs through the OpenMP target runtime, including across multiple nodes using MPI. However, performance is still limited compared to a CPU indicating that GPU support is functional but not yet well optimised. This work serves as a demonstration of current functionality and establishes a baseline for future improvements in PSyclone.

2 Building NEMO

2.1 Manual Compilation

The scripts for manually building NEMO are provided in a Git repository [3]. Detailed steps to compile NEMO are described in `MANUAL_COMPILATION.md` present in the root of the repository.

Even though it is referred as a manual build, the process still relies on Spack to install and manage dependencies (e.g. NetCDF, HDF5), as these are not available prebuilt across all three systems. NEMO itself is built manually.

The manual build scripts use an older method to transform the source files through PSyclone. Instead of using the newer `psyclonefc` wrapper, PSyclone is invoked indirectly via `sct_psyclone.sh`. Modified `sct_psyclone.sh` and `makenemo` are provided in the `etc` directory to facilitate this. This method is not compatible with more recent versions of NEMO (e.g. v5.1), where the `maknemo` has changed significantly. Hence, the Spack package is recommended if you are interested in using the latest branch of NEMO.

2.2 Spack Package

A Spack package is provided for building NEMO [1], which uses the `psyclonefc` wrapper instead of the older method used in manual compilation.

System-specific Spack configuration files for Archer 2, Isambard 3, and Isambard-AI are provided in the `benchmark-nemo` repository under the `spack-configs` directory. The results presented in this report can be reproduced by copying the appropriate configuration files into the local `.spack` directory on the target system.

Most features are currently supported through the Spack package, except XIOS, which is broken upstream at the time of benchmarking. OpenMP (CPU) and GPU support depends on the PSyclone spack package.

Once installed and loaded into the spack environment, the package provides a `nemo-wrapper` command in the user `PATH`. This wrapper is used to generate a local run directory containing a `nemo` executable. Generating the run directory allows flexibility, as users can modify runtime inputs like namelists directly within the run directory if required.

3 State of GPU Support

GPU support in NEMO is currently enabled through OpenMP offloading. Source files are passed to PSyclone which transforms the code to add OpenMP pragmas and other optimisations that make data movement GPU-friendly. At present, some components remain unsupported, notably Sea-Ice and XIOS.

Work is ongoing to address limitations in the Sea-Ice gather–scatter patterns, which are not well suited for GPU memory accesses. But this has not yet been added to PSyclone upstream.

XIOS consists of two components: a server and a client library. Within NEMO, data is first aggregated before calling the XIOS client routines. This aggregation step currently runs on a single core and acts as a bottleneck for GPU performance. XIOS client itself has no GPU support yet.

Work is also underway to improve parallelism on GPUs. For example, loops in subroutines such as `zdftke` are now transformed to expose more parallelism, reversing changes introduced between NEMO v4.2 and v5 that prioritised cache locality on CPUs.

It is currently assumed that data movement between the CPU and GPU is handled automatically. At present, NEMO requires Nvidia GPUs, where both unified and managed memory have been confirmed to work. Support for AMD GPUs is limited due to compiler issues, although work is ongoing on the vendor side to address these. NEMO does not currently run on Intel GPUs.

Even on Nvidia systems, compiler support is limited. So far, only `nvfortran` has been successfully used, and even then with a number of issues. On Isambard-AI, additional source files had to be excluded from the PSyclone transformation due to repeated compiler failures (signal 11).

So overall, running NEMO on GPUs is now possible. But significant effort is still required, mostly from vendors and compilers, before it can be considered stable and performant on GPUs.

4 Benchmark Setup

System and compiler details are provided in Section 5. All benchmarks are run on the BENCH configuration with Sea-Ice and XIOS disabled.

`run.stat` is generated for checking correctness and is enabled either by setting `sn_cfctl%l_runstat = .true.` in the `namctl` namelist, or by using the `+stats` variant in the Spack package.

For GPU runs on Isambard-AI, offloading is enabled through Spack using `+omp +omp_offload` with `omp_flags="-mp=gpu -gpu=mem:unified:nomanaged"`. PSyclone is pinned to commit `c9a794c`, the latest at the time of benchmarking. Additionally, the file `sbcfwb.f90` is added to offload exclusions in PSyclone due to compiler failing with a signal 11.

5 System Details

The benchmarks in this report were carried out on three HPC systems across the UK: Archer 2, Isambard 3, and Isambard-AI. Details of each system are listed below.

Archer 2:

* CPU: AMD EPYC 7742, 2.25 GHz, 64-core
* Cores per node: 128
* Memory per node: 256 GiB
* Interconnect: HPE Cray Slingshot 10
* Compiler used: CCE 16.0.1

Isambard 3:

* CPU: Nvidia Grace Superchip 2x72-core
* Cores per node: 144
* Memory per node: 240 GiB
* Interconnect: HPE Cray Slingshot 11
* Compiler used: NVHPC 24.3

Isambard-AI:

* CPU: Nvidia Grace 72-core
* GPU: Nvidia GH200
* Cores per node: 288
* Memory per node: 480 GiB
* GPUs per node: 4
* GPU memory per node: 384 GiB
* Interconnect: HPE Cray Slingshot 11
* Compiler used: NVHPC 24.11

6 Scaling Results

Figures 1 and 2 show the scaling results for the ORCA1 and ORCA025 resolution models. The ORCA resolutions scale globally in two spatial dimensions, with the vertical dimension held constant. As a result, both computational work and memory requirements scale quadratically with resolution. In practice, this means ORCA025 is approximately 16× larger than ORCA1, while ORCA12 and ORCA36 are each 9× larger than the preceding resolution.

Performance is measured in simulated years per day (SYPD). For each system, MPI process counts are chosen to fully utilise a node: 128 processes on ARCHER2, 144 on Isambard 3, and 4 processes (one per GPU) on Isambard-AI. The number of timesteps is set to 200 for all runs in order to keep the computational cost manageable.

The scaling results for these lower-resolutions are relatively uneventful, as the problem sizes are too small to effectively utilise the available hardware. In particular, GPU runs are dominated by kernel launch overheads.

In terms of performance, Isambard 3 significantly outperforms Isambard-AI, achieving approximately 38× higher throughput for ORCA1 and around 8× for ORCA025. Similarly, Archer 2 out-

performs Isambard-AI by around 10× for ORCA1 and 2.7× for ORCA025. Isambard 3 achieves the highest performance mainly due to optimisations introduced in NEMO v5, which improve cache utilisation, reaching approximately 98% efficiency in both L1 and L2 caches on Nvidia Grace CPU. It is important to note that this comparison considers a single Grace superchip on Isambard 3 versus four GH200 GPUs per node on Isambard-AI.

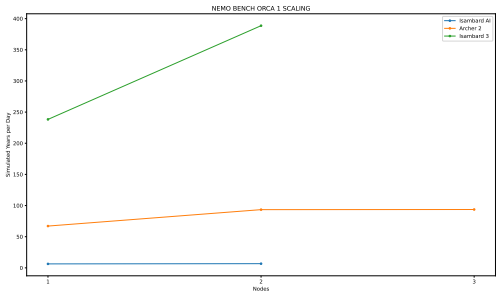

Figure 1: ORCA1 BENCH

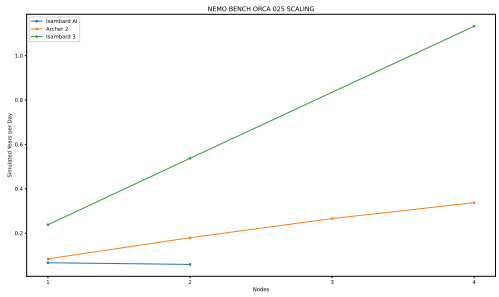

Figure 2: ORCA025 BENCH

Figures 3 and 4 show the scaling results for the ORCA12 and ORCA36 resolutions. Larger resolutions on Isambard 3 require additional nodes due to memory constraints, as each node contains only one Nvidia Grace CPU. For the ORCA12 resolution, using a comparable number of nodes i.e. 4, Isambard 3 achieves approximately 5.7× the performance of Isambard-AI, while Archer 2 reaches around 1.9×. The performance gap is smaller than in the lower-resolution cases because the increased computation on the GPU reduces the relative impact of kernel launch overheads.

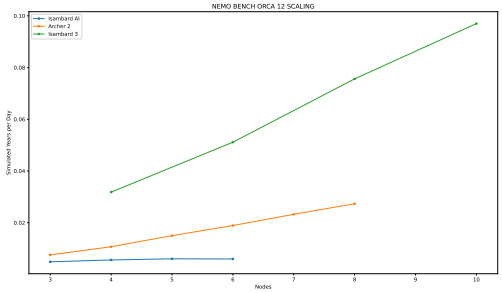

Figure 3: ORCA12 BENCH

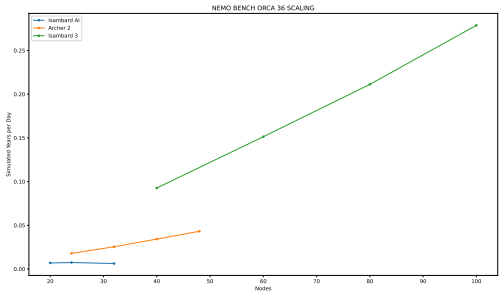

Figure 4: ORCA36 BENCH

For the largest resolution ORCA36, Archer 2 is 6.9× and Isambard 3 is 44.4× faster than Isambard-AI on the highest benchmarked node count. The reason for this performance drop is the bottleneck at `lbc_lnk`. `lbc_lnk` is responsible for data packing and halo exchanges, and it is been observed as a bottleneck on CPUs as well. But on GPUs, it scales extremely poorly. On 20 nodes (80 GPUs), it takes 4351s (58%) of the runtime. As we scale up to 48 nodes, the bottleneck increases and this subroutine now takes 12659s (85%) of the runtime. It was initially suspected that this is because `lbc_lnk` uses derived types containing pointers in order to pack multiple halo exchanges together. Alexey from Barcelona Supercomputing Center has re-written this to use a 1D array with index arithmetic but still runs into the same issue. Currently, this is a known issue but no fix has been identified yet. This implies there are bottlenecks for scaling of NEMOv5 on GPUs, and hence the GPU port is not considered ready for production use.

7 Conclusion

NEMOv5 now runs on Nvidia GPUs. It also supports running on multiple GPUs and can be offloaded safely with minor considerations. There

are still bottlenecks that prevent it to achieve optimal performance and it should considered under heavy development. Future performance improvements will be added directly to PSyclone without modifying the NEMO source code.

8 Acknowledgments

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References

[1] Aditya Sadawarte at. al. *spack-packages repository*. 2025. URL: https://github.com/ukri-bench/spack-packages.

[2] R. Ford et al. *PSyclone (3.1)*. 2025. DOI: 10.5281/zenodo.14925500. URL: https://doi.org/10.5281/zenodo.14925500.

[3] Aditya Sadawarte, Tom Deakin, and Garth N. Wells. *benchmark-nemo repository*. 2025. URL: https://github.com/ukri-bench/benchmark-nemo.